

BAAZ & UDAY JKBOSE

2026 CLASS 11TH



ANIMAL KINGDOM

LECTURE-5

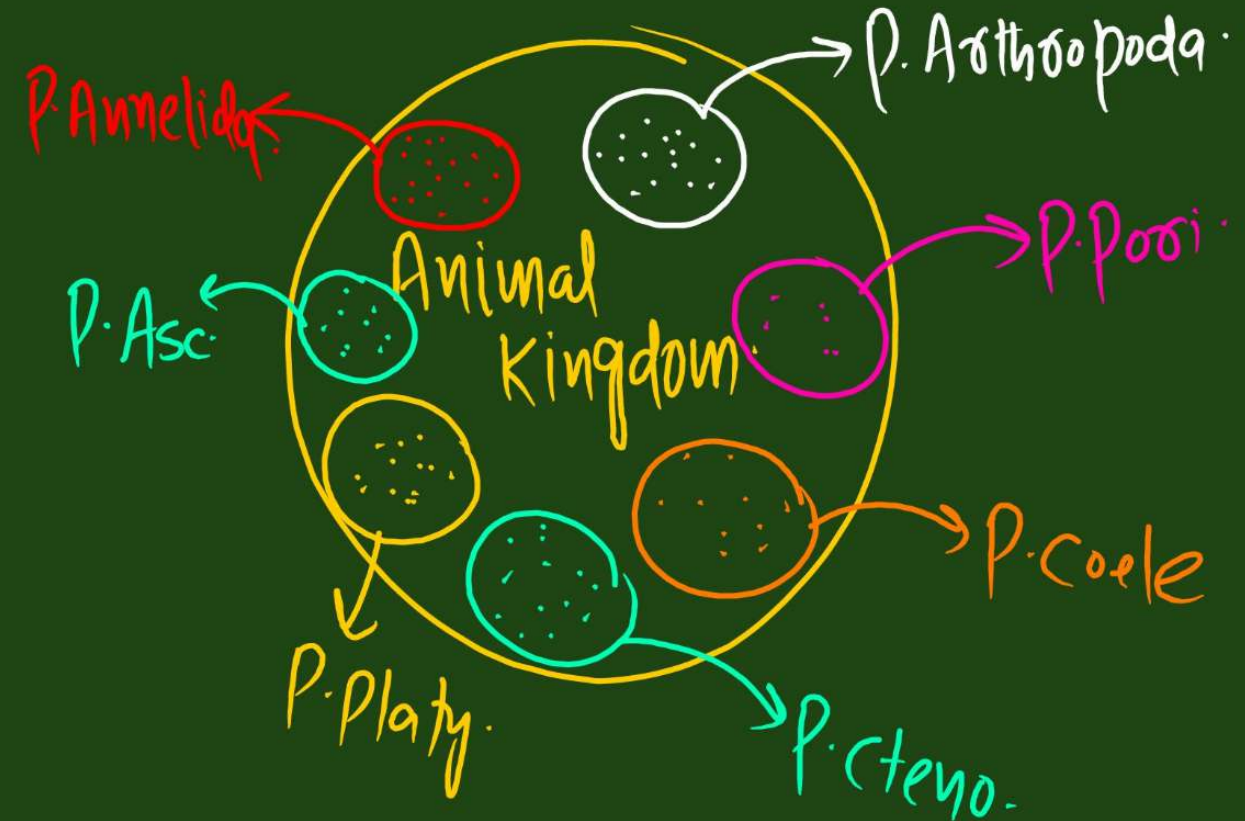
ZOOLOGY

DAVOOD SIR (CEO of NG)



Topics To Be Covered

- 1 **Phylum Annelida**
- 2 **Phylum Arthropoda**
- 3 **Previous Year NEET MCQ's**

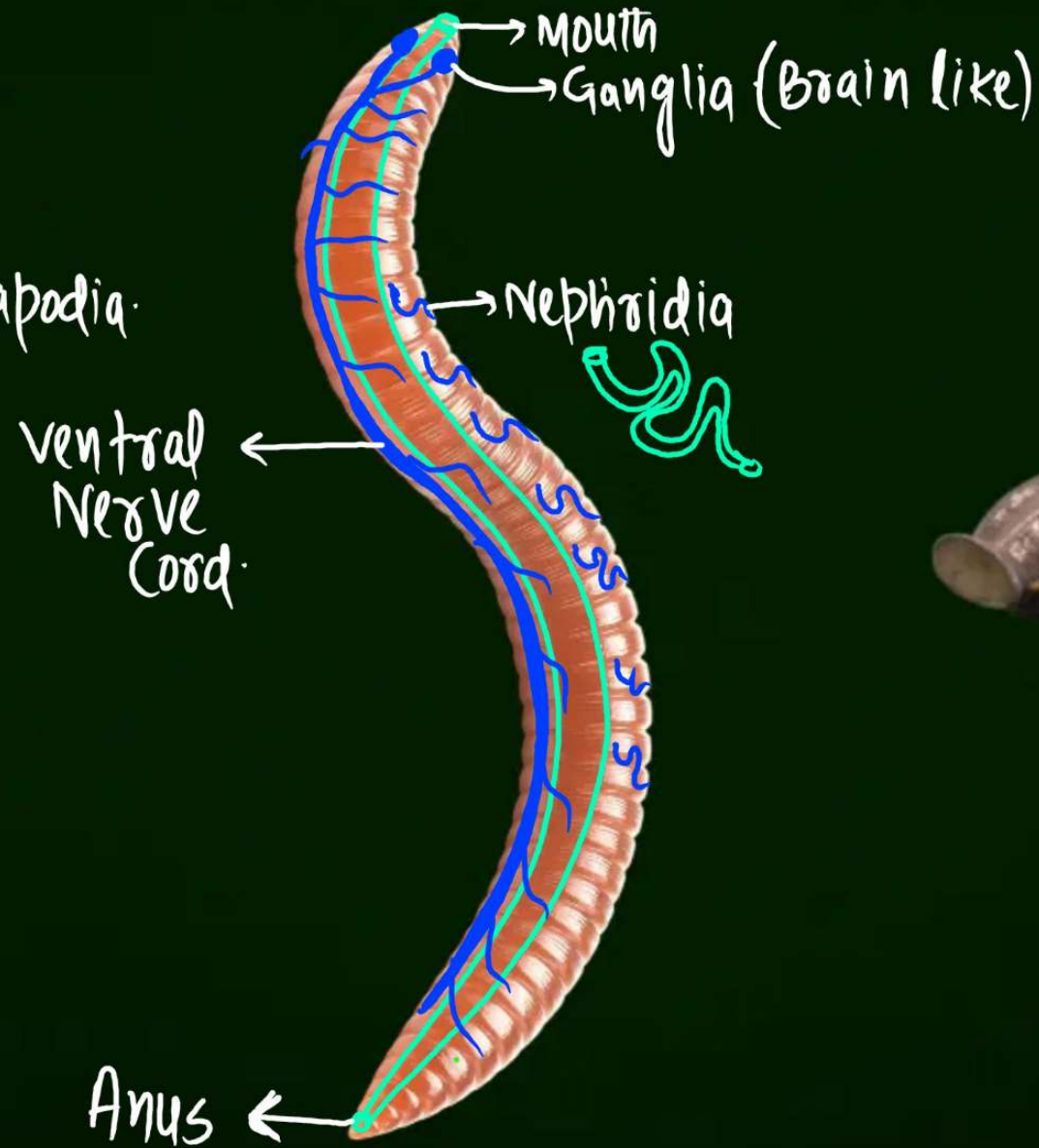


Phylum Annelida (The Segmented Worms)

Neries (Aquatic)



Earthworm (Terrestrial)



Phylum Annelida (The Segmented Worms)

Annelida (Latin word) → means a small Ring → ○

Habitat: Aquatic (Marine + Fresh water), Terrestrial, Parasites

Body Structure: Bilateral symmetry, Triploblastic, Coelomates, organ system level of organisation.

Segmentation: → Body is divided into segments/Metameres (Metameric segmentation)

Circulatory system: Closed type.

Body and Movement:

→ they have longitudinal muscles & circular muscles for movement.

Excretory system:

→ Aquatic Annelida (Neries) have Parapodia (Lateral Appendages) for swimming.

Nervous System:

→ they have Nephridia for excretion & osmoregulation.

→ Nervous system is made of Ganglia (Group of Neurons). Ganglia is connected with a ventral nerve cord (like spinal cord).

Phylum Annelida (The Segmented Worms)

Reproduction: they show sexual reproduction.
 ↳ Some members are dioecious (Nereis)
 ↳ Some are monoecious (Earthworm + leech)
 ↳ Both internal & external fertilization with direct development except Nereis.

Examples:

- *Nereis* (sandworm), *Pheretima* (Earthworm), *Hirudinaria* (Blood-sucking leech)

• UNIQUE FEATURES:-

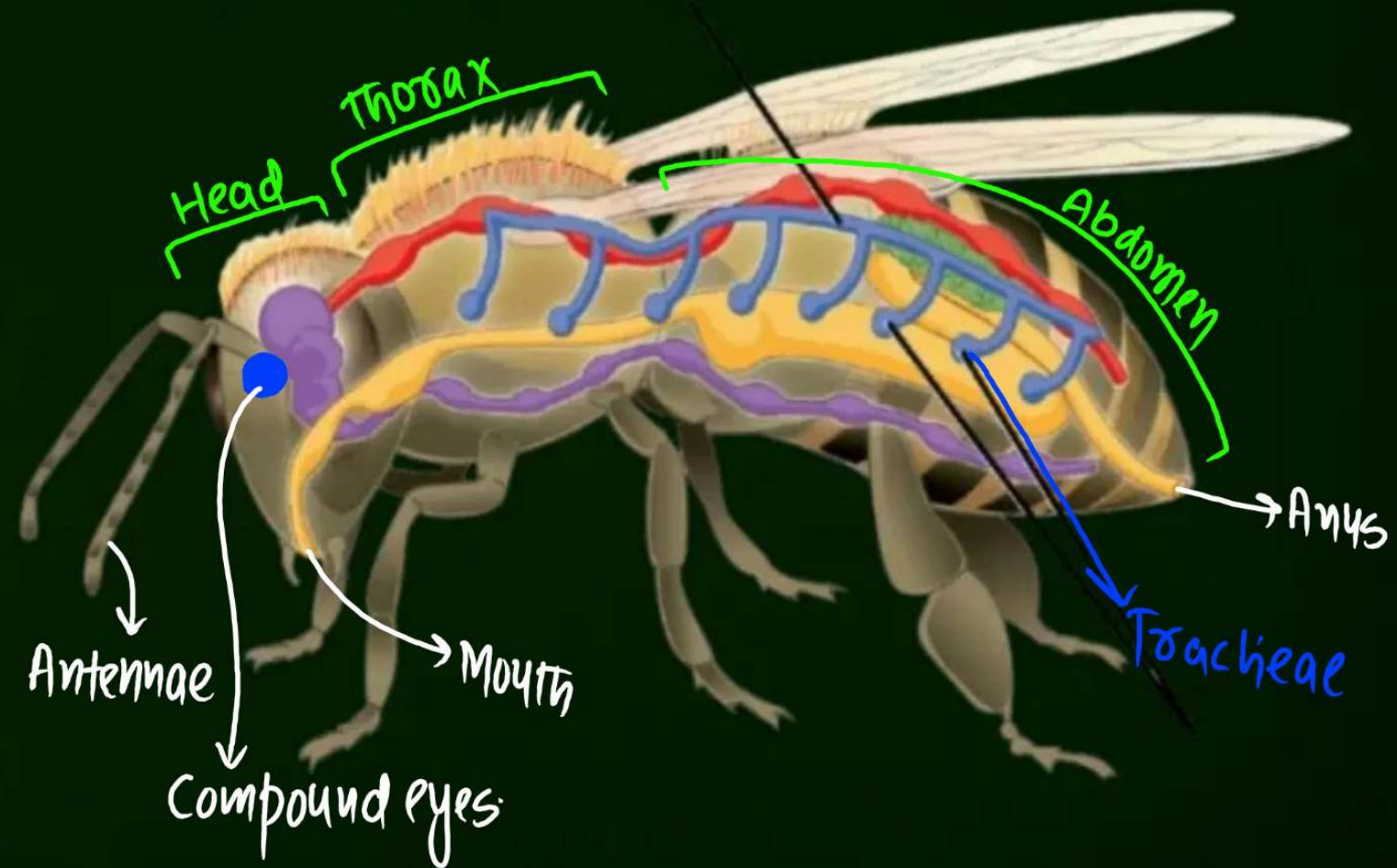
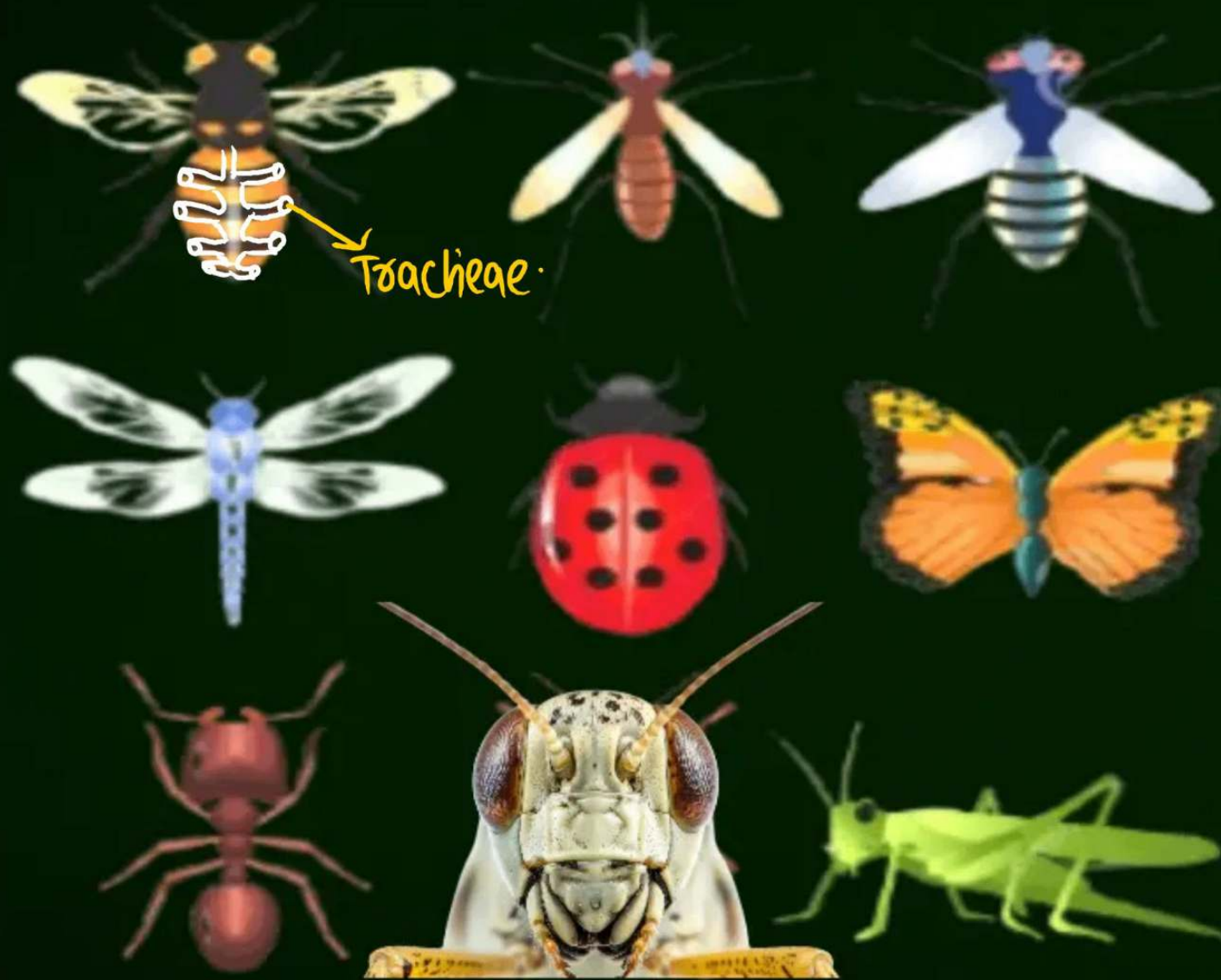
- Metameric segmentation.
- Nephridia for excretion.

Phylum Annelida (The Segmented Worms)

"A SEGMENTED COEL worm MOVES with PARAPODIA and EXCRETES by NEPHRIDIA."

<i>A SEGMENTED</i>	→ <i>Body is segmented (metameres)</i>
<i>COEL</i>	→ <i>Coelomate (true body cavity)</i>
<i>worm</i>	→ <i>Common annelids are worms (Earthworm, Leech, Nereis)</i>
<i>MOVES</i>	→ <i>Movement by longitudinal and circular muscles</i>
<i>PARAPODIA</i>	→ <i>Aquatic forms (like Nereis) use parapodia to swim</i>
<i>EXCRETES</i>	→ <i>Excretory organ is nephridia</i>
<i>NEPHRIDIA</i>	→ <i>Used for excretion and osmoregulation</i>

Phylum Arthropoda (Joint Legged animals)



Phylum Arthropoda (Joint Legged animals) → insects.

- Arthro → joints & poda → Appendages.
- It is the Largest phylum with $\frac{2}{3}$ (9-lakh) species.

Body features:- Bilateral symmetry, Triploblastic, Haemocoel, organ system level of organisation and open circulatory system.

Habitat: → On Land, In water and also as parasites.

- ↳ Body is covered with chitinous Exoskeleton
- ↳ Body is divided into three parts

a) Head: Have sensory organs like Antennae, Compound eyes and statocysts for balance.

b) Thorax: jointed appendages arise from thorax.

c) Abdomen:

Respiratory System: Gills, Book Gills in Aquatic insects and tracheae or book lungs in terrestrial insects

Excretory System: Malpighian tubules or green glands for excretion

Reproduction: Mostly dioecious with internal fertilization and lay eggs (oviparous).
Direct as well as indirect development (Metamorphosis).

Phylum Arthropoda (Joint Legged animals)

Examples:

Economically important insects: Apis (Honey bee), Bombyx (Silkworm), Laccifer (Lac insect)

Vectors: Anopheles, Culex, Aedes (Mosquitoes)

Gregarious pest- Locusta (Locust)

Living fossil: Limulus (King crab)

Unique features

- ① jointed appendages.
- ② Chitinous exoskeleton
- ③ Compound eyes.
- ④ Hemocoel.
- ⑤ Malpighian tubules.

Kaun Banayga Nabkaar: NEET-2025

Exoskeleton of arthropods is composed of

- A** Cutin
- B** Cellulose
- C** Chitin ✓
- D** Glucosamine



Thank
You